

CHOI WON JIP

Ship Design & Manufacturing DX Engineer | Engineering Systems / CAD-PDM-PLM Support

Portfolio: <https://jiwonjae-portfolio.vercel.app/>

PROFESSIONAL SUMMARY

Ship design engineer with 4+ years of professional experience in structural design, complex drawing analysis, construction workflow coordination, and schedule recovery. Experienced in translating engineering constraints into practical priorities and coordinating responses when equipment, weather, or resource conditions threaten delivery.

Combines shipbuilding-domain experience with independently developed capabilities in TypeScript, Next.js, PostgreSQL, controlled drawing-revision workflows, access control, audit logging, AI-assisted knowledge retrieval, and software testing. DrawingFlow and DocuMind are independent DX case studies, not systems deployed at Manmul Joseon.

Seeking shipbuilding or manufacturing DX, engineering systems, CAD/PDM/PLM application support, technical operations, workflow automation, or quality/process-improvement roles where domain knowledge and digital implementation can be applied together.

PROFESSIONAL EXPERIENCE

Manmul Joseon - Ship Design / Engineering

Busan, Korea | Sep 2021 - Present | Ship design, construction, and engineering | 30 employees | 10-person project team

- Supported structural ship design across basic design, detailed design, and construction-process coordination.
- Interpreted complex ship-structure drawings and translated design constraints into practical construction actions.
- Identified bottlenecks across remaining construction work and reprioritized schedules, personnel, and equipment around critical tasks.
- When equipment failures and adverse weather threatened the schedule, rebuilt the execution sequence and resource plan, completing the project without a delivery delay.
- Responded to uncertain field conditions with engineering-based design adjustments, clear documentation, and coordination with project stakeholders.
- Developed a disciplined approach to specification reading, issue diagnosis, change communication, and responsibility for operational outcomes.

DOMAIN AND DIGITAL CAPABILITIES

Shipbuilding / Engineering Structural drawing analysis, basic and detailed design support, construction sequencing, bottleneck diagnosis	Process Improvement Workflow mapping, constraint analysis, schedule and resource reprioritization, change documentation
Engineering Systems Drawing revision control, impact tracking, release gates, document traceability, project RBAC, append-only audit logs	Software TypeScript, JavaScript, React, Next.js, PostgreSQL, Prisma, pgvector, Auth.js, REST-style APIs
Quality / Automation Vitest, Playwright, input validation, regression checks, DB integrity verification, documented limitations	AI-assisted Tools OpenAI API, RAG, embeddings, semantic search, source-cited answers, human-reviewable outputs

INDEPENDENT DX AND INTERNAL-TOOL PROJECTS

DrawingFlow - Drawing Revision Impact Tracker

Next.js, TypeScript, PostgreSQL, Prisma, Auth.js, Zod, Vitest, Playwright | drawing-revision-impact-tracker.vercel.app

- Built and deployed a drawing-revision workflow based on ship-design and manufacturing handoffs: Draft -> In Review -> Approved -> Issued -> Closed.
- Enforced independent approval, effective-date issue gates, production acknowledgement, project-scoped roles, and an append-only PostgreSQL audit trail.
- Added downstream impact registration, a risk-ordered review queue, validated CSV drawing-register import, a typed REST PDM adapter boundary, and a durable webhook notification outbox.
- Verified workflow and integration rules with 12 unit/integration tests, four multi-role Playwright scenarios, and a database-level audit immutability check.
- Uses synthetic data only; it does not claim employer deployment or a proprietary CAD/PDM connection.

DocuMind - Engineering Knowledge and Traceability Case Study

Next.js, TypeScript, PostgreSQL, Prisma, pgvector, Auth.js, OpenAI API, Vitest

- Built an independent internal knowledge workflow system with authenticated document ingestion, semantic retrieval, and source-cited AI answers.
- Designed owner-scoped access control and audit logging to make document use traceable and reviewable.
- Added validation and release-gate coverage across ingestion, retrieval, citation formatting, access boundaries, and API behavior.
- Maintains 55 Vitest files / 339 automated tests and documentation separating implemented features, limitations, and future work.
- Relevant manufacturing application: controlled retrieval of technical specifications and project documents, with project-level access and visible source evidence.

SVG Converter - Browser-Based Technical File Utility

TypeScript, Canvas API, SVG

- Built a browser-side file-processing utility with clear input validation and deterministic output generation.
- Improved image-processing performance and reduced generated SVG output size.
- Demonstrates practical handling of technical files, edge cases, and user-verifiable output.

EDUCATION AND CREDENTIALS

- Changwon National University - B.E., Naval Architecture and Ocean Engineering | Mar 2015 - Sep 2021
- Korean Engineer Information Processing | Jul 2020
- TOEIC Speaking 140 | Apr 2022
- Korean Class 1 Driver's License | Feb 2015

LANGUAGES AND RELOCATION

- Korean: Native
- Japanese: Conversational; working holiday experience
- English: Technical reading/writing
- Japan relocation intended; visa sponsorship required

TARGET ROLES

Ship Design / Marine Engineering | Manufacturing or Shipbuilding DX | Engineering Systems | CAD/PDM/PLM Application Support | Technical Operations | Workflow Automation | Quality and Process Improvement